

● HARD

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Geometry and Trigonometry

10

10 answers · Rationale-rich · Teach from doc

Q1**D****D.** 1,768

Choice D is correct. It's given that the radius of the circle is 168 millimeters. Since points M and N both lie on the circle, segments GM and GN are both radii. Therefore, segments GM and GN each have length 168 millimeters. Two segments that are tangent to a circle and have a common exterior endpoint have equal length. Therefore, segment MH and segment NH have equal length. Let x represent the length of segment MH . Then x also represents the length of segment NH . It's given that the perimeter of quadrilateral $GMHN$ is 3,856 millimeters. Since the perimeter of a quadrilateral is equal to the sum of the lengths of the sides of the quadrilateral, $3,856 = 168 + 168 + x + x$, or $3,856 = 336 + 2x$. Subtracting 336 from both sides of this equation yields $3,520 = 2x$, and dividing both sides of this equation by 2 yields $1,760 = x$. Therefore, the length of segment MH is 1,760 millimeters. A line segment that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. Therefore, segment GM is perpendicular to segment MH . Since perpendicular segments form right angles, angle GMH is a right angle. Therefore, triangle GMH is a right triangle with legs of length 1,760 millimeters and 168 millimeters, and hypotenuse GH . By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting 1,760 for a and 168 for b in this equation yields $1,760^2 + 168^2 = c^2$, or $3,125,824 = c^2$. Taking the square root of both sides of this equation yields $\pm 1,768 = c$. Since c represents a length, which must be positive, the value of c is 1,768. Therefore, the length of segment GH is 1,768 millimeters, so the distance between points G and H is 1,768 millimeters.

Choice A is incorrect. This is the distance between points G and M and between points G and N , not the distance between points G and H .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the distance between points M and H and between points N and H , not the distance between points G and H .

Q2**D****D.** $\frac{55}{48}$

Choice D is correct. The hypotenuse of triangle RST is the longest side and is across from the right angle. The longest side length given is 584, which is the length of side TR . Therefore, the hypotenuse of triangle RST is side TR , so the right angle is angle S . The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side, which is the side across from the angle, to the length of the adjacent side, which is the side closest to the angle that is not the hypotenuse. It follows that the opposite side of angle T is side RS and the adjacent side of angle T is side ST . Therefore, $\tan T = \frac{RS}{ST}$. Substituting 440 for RS and 384 for ST in this equation yields $\tan T = \frac{440}{384}$. This is equivalent to $\tan T = \frac{55}{48}$. It's given that triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . It follows that R corresponds to U . Therefore, the hypotenuse of triangle UVW is side WU , which means $\tan W = \frac{UV}{VW}$. Since the lengths of corresponding sides of similar triangles are proportional, $\frac{RS}{ST} = \frac{UV}{VW}$. Therefore, $\tan W = \frac{UV}{VW}$ is equivalent to $\tan W = \frac{RS}{ST}$, or $\tan W = \tan T$. Thus, $\tan W = \frac{55}{48}$.

Choice A is incorrect. This is the value of $\cos W$, not $\tan W$.

Choice B is incorrect. This is the value of $\sin W$, not $\tan W$.

Choice C is incorrect. This is the value of $\frac{1}{\tan W}$, not $\tan W$.

Q3**D****D.** $\frac{38}{15}$

Choice D is correct. It's given that in triangle ABC , the measure of angle B is 90° and BD is an altitude of the triangle. Therefore, the measure of angle BDC is 90° . It follows that angle B is congruent to angle D and angle C is congruent to angle C . By the angle-angle similarity postulate, triangle ABC is similar to triangle BDC . Since triangles ABC and BDC are similar, it follows that $\frac{AC}{AB} = \frac{BC}{BD}$. It's also given that the length of $\bar{AB}$ is 15 and the length of $\bar{AC}$ is 23 greater than the length of $\bar{AB}$. Therefore, the length of $\bar{AC}$ is $15 + 23$, or 38. Substituting 15 for AB and 38 for AC in the equation $\frac{AC}{AB} = \frac{BC}{BD}$ yields $\frac{38}{15} = \frac{BC}{BD}$. Therefore, the value of $\frac{BC}{BD}$ is $\frac{38}{15}$.

Choice A is incorrect. This is the value of $\frac{BD}{BC}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4**D****D.** 41,200

Choice D is correct. The volume, V , of a sphere is given by $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. Since a hemisphere is half of a sphere, it follows that the volume, V , of a hemisphere is given by $V = \frac{1}{2} \cdot \frac{4}{3}\pi r^3$, or $V = \frac{2}{3}\pi r^3$. Substituting 27 for r in this formula yields $V = \frac{2}{3}\pi 27^3$, which gives $V = 13,122\pi$, or V is approximately equal to 41,223.98. Therefore, the choice that is closest to the volume, in cubic inches, of this hemisphere is 41,200.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q5**B**

B. $(x - 19)^2 + (y - 18)^2 = 81k^2$

Choice B is correct. The equation of a circle in the xy -plane can be written as

$(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . It's given that this circle has a center at $(19, 18)$ and a radius of $9k$. Substituting 19 for h , 18 for k , and $9k$ for r in $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 19)^2 + (y - 18)^2 = (9k)^2$, or $(x - 19)^2 + (y - 18)^2 = 81k^2$. Therefore, the equation that represents this circle is $(x - 19)^2 + (y - 18)^2 = 81k^2$.

Choice A is incorrect. This equation represents a circle with radius $9\sqrt{k}$, not $9k$.

Choice C is incorrect. This equation represents a circle with radius $\sqrt{9k}$, not $9k$.

Choice D is incorrect. This equation represents a circle with radius $\sqrt{9} k$, not $9k$.

Q6**113**

The correct answer is 113. It's given that the legs of a right triangle have lengths 24 centimeters and 21 centimeters. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. It follows that if h represents the length, in centimeters, of the hypotenuse of the right triangle, $h^2 = 24^2 + 21^2$. This equation is equivalent to $h^2 = 1,017$. Taking the square root of each side of this equation yields $h = \sqrt{1,017}$. This equation can be rewritten as $h = \sqrt{9 \cdot 113}$, or $h = \sqrt{9} \cdot \sqrt{113}$. This equation is equivalent to $h = 3\sqrt{113}$. It's given that the length of the triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$. It follows that the value of d is 113.

Q7**C**

C. 2.4

Choice C is correct. First, $\overline{MP}$ is the hypotenuse of right $\triangle MNP$, whose legs have lengths 3 and 4. Therefore, $(MP)^2 = 3^2 + 4^2$, so $(MP)^2 = 25$ and $MP = 5$. Second, because $\angle MNP$ corresponds to $\angle NQP$ and because $\angle MPN$ corresponds to $\angle NPQ$, $\triangle MNP$ is similar to $\triangle NQP$. The ratio of corresponding sides of similar triangles is constant, so $\frac{NQ}{MN} = \frac{NP}{MP}$. Since $MP = 5$ and it's given that $MN = 3$ and $NP = 4$, $\frac{NQ}{3} = \frac{4}{5}$. Solving for NQ results in $NQ = \frac{12}{5}$, or 2.4.

Choices A, B, and D are incorrect and may result from setting up incorrect ratios.

Q8**66**

The correct answer is 66. It's given that the right circular cone has a height of 22 centimeters cm and a base with a diameter of 6 cm. Since the diameter of the base of the cone is 6 cm, the radius of the base is 3 cm. The volume V , in cm^3 , of a right circular cone can be found using the formula $V = \frac{1}{3}\pi r^2 h$, where h is the height, in cm, and r is the radius, in cm, of the base of the cone. Substituting 22 for h and 3 for r in this formula yields $V = \frac{1}{3}\pi 3^2 22$, or $V = 66\pi$. Therefore, the volume of the cone is $66\pi \text{ cm}^3$. It's given that the volume of the cone is $n\pi \text{ cm}^3$. Therefore, the value of n is 66.

Q9**B**B. $(-10, -8)$

Choice B is correct. The standard equation of a circle in the xy -plane is of the form $(x-h)^2 + (y-k)^2 = r^2$, where (h, k) are the coordinates of the center of the circle and r is the radius. The given equation can be rewritten in standard form by completing the squares. So the sum of the first two terms, $x^2 + 20x$, needs a 100 to complete the square, and the sum of the second two terms, $y^2 + 16y$, needs a 64 to complete the square. Adding 100 and 64 to both sides of the given equation yields $(x^2 + 20x + 100) + (y^2 + 16y + 64) = -20 + 100 + 64$, which is equivalent to $(x + 10)^2 + (y + 8)^2 = 144$. Therefore, the coordinates of the center of the circle are $(-10, -8)$.

Choices A, C, and D are incorrect and may result from computational errors made when attempting to complete the squares or when identifying the coordinates of the center.

Q10**0**

The correct answer is 0. Note that no matter where point W is on $\overline{RT}$, the sum of the measures of $\angle RSW$ and $\angle WST$ is equal to the measure of $\angle RST$, which is 90° . Thus, $\angle RSW$ and $\angle WST$ are complementary angles. Since the cosine of an angle is equal to the sine of its complementary angle, $\cos(\angle RSW) = \sin(\angle WST)$. Therefore, $\cos(\angle RSW) - \sin(\angle WST) = 0$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank